

PROCESS FOR THE TREATMENT OF INDUSTRIAL WASTEWATER

Industrial wastewater treatment involves customized physical, chemical, and biological processes designed to remove toxic pollutants from manufacturing byproducts. It allows factories to safely recycle water or discharge it into municipal sewers or natural bodies of water without violating environmental regulations.

Because industrial effluent varies heavily based on the specific industry (e.g., textiles, pharmaceuticals, petrochemicals), treatment is typically carried out in customized multi-stage systems called Effluent Treatment Plants (ETPs).

Source of effluent

Effluent is treated or untreated liquid waste that flows out of a facility or system into a natural body of water or the ground. It primarily originates from three distinct sectors: domestic households, industrial manufacturing, and agricultural operations.

1. Domestic and Municipal Sources

- **Sewage & Wastewater:** Includes water from toilets (black water), sinks, baths, and appliances (grey water).
- **Runoff:** Municipalities collect this waste in sewage treatment plants to extract harmful microbes, organic matter, and household chemicals before releasing the treated effluent.
- **Septic Systems:** In rural areas, on-site decentralized septic tanks discharge effluent into the surrounding soil.

2. Industrial Facilities

- **Manufacturing Plants:** Factories (such as those producing textiles, leather, electronics, or chemicals) discharge heavy metals, toxic solvents, and dyes.
- **Power Plants & Mining:** Mining operations release minerals, while thermal power stations discharge heated cooling water into the environment.
- **Food Processing:** Generates highly organic, biodegradable waste that can rapidly deplete oxygen levels in local waterways.

3. Agricultural Systems

- **Nutrient Runoff:** Fields fertilized with chemicals (releasing nitrogen and phosphorus) or pesticides produce contaminated agricultural runoff during heavy rainfall.

- **Livestock Waste:** Large-scale farming setups like dairy sheds and piggeries produce animal waste-heavy effluent that requires targeted filtration.

Discharge requirements

According to the general requirements, it is stipulated that in all cases, liquid industrial discharges must have the following characteristics

- a. pH between 5,5 and 8,5 (if the effluent is neutralized by lime, the pH may be between 5,5 and 9,5)
- b. Temperature less than or at most equal to 30°C
- c. Absence of hydroxylated cyclic compounds from the phenol series and their halogen derivatives
- d. Absence of any coloring substance or substance likely to give off foul odors and abnormal flavors
- e. Absence of any substance likely to ignite directly or indirectly after mixing with other effluents, or to release toxic vapors
- f. Absence of floating materials or any substance harmful to the proper functioning of the evacuation and treatment works
- g. No substances that may affect aquatic wildlife downstream of the point of discharge
- h. No risk of radiation to the neighborhood
- i. There are also additional requirements for physical criteria, such as conductivity, decantable materials in cm³/L in 2 h
- j. Chemical or biochemical criteria such as nitrogen and dissolved oxygen in mg/L, BOD₅ in mg/L, and oxidability to potassium permanganate in mg/L.

It should be added that some wastewater has high concentrations of phosphates. Phosphorus removal is essential to prevent eutrophication of lakes.

Methods of industrial wastewater treatment

Industrial wastewater treatment plants use several different methods to treat wastewater. The exact methods used depend on your industry and the specific byproducts in the water.

Some common treatment methods for industrial wastewater include:

- **Physical treatment:** These treatment methods physically remove solids from the water using filtration, screening, and other similar methods. This can also include membrane

filtration, which uses techniques like reverse osmosis to remove contaminants that have been suspended in water.

- **Chemical treatment:** Chemical treatment methods are used to remove specific contaminants or neutralize the pH. They create chemical reactions within the water to change its composition. Some forms of chemical treatment include oxidation, ion exchange, and pH adjustment.
- **Biological treatment:** These treatment methods are used to break down and neutralize organic compounds that are present in the water. This is often done using activated sludge and other microorganisms, which convert the organic materials into carbon dioxide or methane.

Physical water treatment

Processes used for the treatment of wastewater in which change is brought about by means of or through the application of physical forces are known as physical unit processes. Because physical unit processes were derived originally from observations of the physical world, they were the first treatment methods to be used. The physical unit processes most commonly used in preliminary and primary treatment of wastewater include some or all of the following (1) screening, (2) coarse solids reduction (comminution, maceration, and screenings grinding), (3) mixing and flocculation, (4) gravity separation, (5) grit removal, (6) primary sedimentation, (7) high-rate clarification, and (8) flotation.

Screening

The first unit process generally encountered in wastewater treatment plants is screening. A screen is a device with openings, generally of uniform size, that is used to retain large solids found in the influent wastewater to the treatment plant or in combined waste water collection systems subject to overflows, especially from stormwater.

The principal role of screening is to remove coarse materials from the flow stream that could (1) damage or clog subsequent process equipment, (2) reduce overall treatment process reliability and effectiveness, or (3) contaminate waterways. Fine screens are sometimes used in place of or following coarse screens where greater removals of solids are required to (1) protect equipment which may be more sensitive to solids such as membrane bioreactors or (2) eliminate materials that may inhibit the beneficial reuse of biosolids.

All aspects of screenings removal, transport, and disposal must be considered in the application of screening devices, including (1) the degree of screenings removal required because of potential effects on downstream treatment processes and equipment, (2) health and safety of the operators as screenings contain pathogenic organisms and attract insects, (3) odor potential, and (4) requirements for handling, transport, prior to disposal (i.e., removal of organics by washing and

reduced water content by pressing), and (5) disposal options. Thus, an integrated approach is required to achieve effective screenings management.

Classification of Screens

Types of screens commonly used in wastewater treatment are identified on Fig. 1. Two general types of screens, coarse and fine screens, are used in preliminary treatment. Fine screens may also be used as an alternative primary treatment process or for removal of additional organic solids from sludge streams prior to sludge processing. Microscreens are used principally for removing residual solids from treated effluents.

Screening elements may consist of parallel bars, rods or wires, grating, wire mesh, or perforated plate, and the openings may be of any shape, but generally are circular or rectangular slots. A screen composed of parallel bars or rods is often called a “bar rack” or a coarse screen and is used for the removal of coarse solids. Fine screens are devices consisting of perforated plates, wedge wire elements, and wire cloth that have smaller openings. The materials removed by these devices are known as screenings.

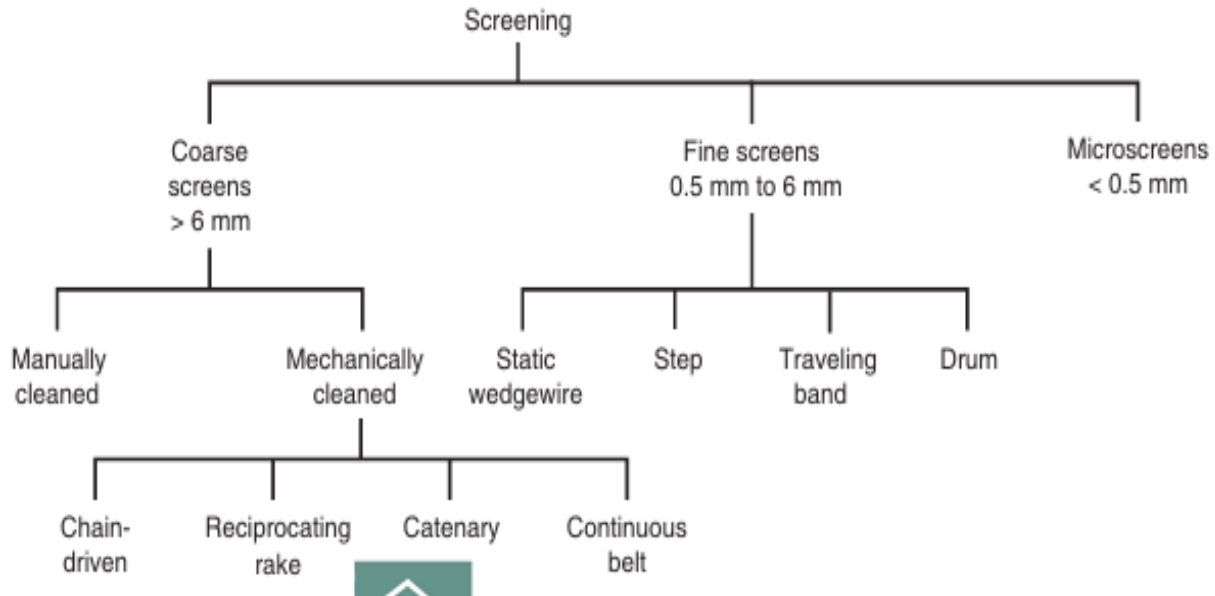


Fig 1: General classification for the types of screens used in wastewater treatment.

Screenings Characteristics and Quantities

Screenings are the material retained on bar racks and screens. The smaller the screen opening, the greater will be the quantity of collected screenings. While no precise definition of screenable material exists, and no recognized method of measuring quantities of screenings is available, screenings exhibit some common properties.

Screenings Retained on Coarse Screens.

Coarse screenings greater than 6 mm (0.25 in.) consist of debris such as rocks, branches, pieces of lumber, leaves, paper, tree roots, plastics, bottles, cans, and rags. Organic matter associated with screenings are also removed with organic content increasing as the spacing decreases. The accumulation of oil and grease can be a serious problem, especially in cold climates. The quantity and characteristics of screenings collected for disposal vary, depending on the type of bar screen, the size of the bar screen opening, the type of sewer system, and the geographic location.

The characteristics of the collection system will also impact the volume of screenings received. Where the collection system has many lift stations with their own screening equipment, the volume of screenings received at the treatment plant will be less. Combined sewer systems typically produce more screenings than separate sanitary sewers and may produce volumes of screenings at up to 20 times the average rate during the “first flush” of peak wet weather flow event.

Screenings Retained on Fine Screens

Fine screenings consist of materials retained on screens with openings of 0.5 mm (0.02 in.) to 6 mm (0.25 in.). The materials retained on fine screens include small rags, paper, plastic materials of various types, razor blades, grit, undecomposed food waste, feces, etc. Compared to coarse screenings, the specific weight of the fine screenings is lower and the moisture content is higher than for coarse screens. Fine screenings contain putrescible matter (including fecal material), substantial amounts of grease and scum and may be or become extremely odorous. Consequently, manual handling of collected fine screenings should be minimized and screening should be conveyed using enclosed conveyors and include grit washing and compacting equipment.

Coarse Screens (Bar Racks)

In wastewater treatment, coarse screens are used to protect pumps, valves, pipelines, and other appurtenances from damage or clogging by rags and large objects. Typically, coarse screens precede fine screens to protect them from damage from large debris. Industrial waste treatment plants may not need coarse screens, depending on the character of the wastes. According to the method used to clean them, coarse screens are designated as either manually cleaned or mechanically cleaned. Screenings, handling, processing, and disposal are discussed following the discussion of the types of screens.

Manually Cleaned Coarse Screens

Manually cleaned coarse screens are used frequently ahead of pumps in small wastewater pumping stations and sometimes at the headworks of small- to medium-sized wastewater treatment plants. Often, they are used for standby screening in bypass channels for service during high-flow periods, when mechanically cleaned screens are being repaired, or in the event of a power failure. Normally, mechanically cleaned screens are provided in lieu of manually cleaned screens to minimize labor required to clean the screens and to reduce flooding resulting from clogging of infrequently cleaned screens.

The screen channel should be designed to prevent the accumulation of grit and other heavy materials in the channel ahead of the screen and following it. The channel floor should be level or should slope downward through the screen without pockets to trap solids. Fillets may be desirable at the base of the sidewalls. The channel preferably should have a straight approach, perpendicular to the bar screen, to promote uniform distribution of screenable solids throughout the flow and on the screen.

Mechanically Cleaned Bar Screens

The design of mechanically cleaned bar screens has evolved over the years to reduce the operating and maintenance problems and to improve the screenings removal capabilities. Many of the newer designs include extensive use of corrosion-resistant materials including stainless steel and plastics. Mechanically cleaned bar screens are divided into four principal types: (1) chain-driven, (2) reciprocating rake, (3) catenary, and (4) continuous belt. Cable-driven bar screens were used extensively in the past but have been replaced in most wastewater applications by the other types of screens. However, cable-driven raking mechanisms are still used in specific applications such as screens ahead of tunnel dewatering pump stations where deep tunnels are used for storage or conveyance of wet weather flow from combined sewer systems.

Chain-Driven Screens

Chain-driven mechanically cleaned bar screens can be divided into categories based on whether the screen is raked to clean from the front (upstream) side or the back (downstream) side and whether the rakes return to the bottom of the bar screen from the front or back. Each type has its advantages and disadvantages, although the general mode of operation is similar. In general, front-cleaned, front-return screens are more efficient in terms of retaining captured solids, but they are less rugged and are susceptible to jamming by solids that collect at the base of the rake. In front-cleaned, back-return screens, the cleaning rakes return to the bottom of the bar screen on the downstream side of the screen, pass under the bottom of the screen, and clean the bar screen as the rake rises. The potential for jamming is minimized, but a hinged plate, which is also subject to jamming, is required to seal the pocket under the screen.

Reciprocating Rake (Climber) Screen.

The reciprocating-rake-type bar screen imitates the movements of a person raking the screen. The rake moves to the base of the screen, engages the bars, and pulls the screenings to the top of the screen, where they are removed. Most screen designs utilize a cogwheel drive mechanism for the rake. The drive motors are either submersible electric or hydraulic type. A major advantage is that all parts requiring maintenance are above the waterline and can be easily inspected and maintained without dewatering the channel. The front-cleaned, front-return feature minimizes solids carryover. The screen uses only one rake instead of multiple rakes that are used with other types of screens. As a result, the reciprocating rake screen may have limited capacity in handling heavy screening loads, particularly in deep channels where a long “reach” is necessary.

Catenary Screen

A catenary screen is a type of front-cleaned, front-return, chain-driven screen, but it has no submerged sprockets. In the catenary screen, the rake is held against the rack by the weight of the chain. If heavy objects become jammed in the bars, the rakes pass over them instead of jamming. The screen, however, has a relatively large “footprint” and requires greater space for installation.

Continuous Belt Screen

The continuous belt screen is a continuous, self-cleaning screening belt that can be used to remove both fine and coarse solids. A large number of screening elements (rakes) are attached to the drive chains; the number of screening elements depends on the depth of the screen channel. Because the screen openings can range from 0.5 to 30 mm (0.02 to 1.18 in.), it can be used as either a fine or a coarse screen. When used as a fine screen, this type of screen is better able to handle coarse solids and upstream protective devices may not be required. Hooks protruding from the belt elements are provided to capture large solids such as cans, sticks, and rags. The screen has no submerged sprocket. Newer designs of internally fed, continuous-belt, rotary screens using wedge wire instead of screen fabric are structurally more rugged.

Fine Screens

The applications for fine screens range over a broad spectrum and include preliminary treatment (following coarse bar screens), primary treatment (as a substitute for primary clarifiers), and treatment of combined sewer overflows. Fine screens can also be used to remove solids from primary effluent that could cause clogging problems in trickling filters or the membranes of membrane bioreactors. They may also be used for removal of solids from sludge flow streams prior to further treatment of the sludge (biosolids). Fine screens used for preliminary treatment are (1) static wedge wire (fixed), (2) rotary drum (3) travelling belt, and (4) step type screens.

Static Wedge Wire Screens

Static wedge wire screens customarily have 0.2 to 1.2 mm (0.01 to 0.06 in.) clear openings and are designed for flowrates of about 400 to 1200 L/m².min (10 to 30 gal/ft².min) of screen area. Headloss ranges from 1.2 to 2 m (4 to 7 ft). The wedge wire medium consists of small, stainless-steel, wedge-shaped bars with the flat part of the wedge facing the flow. Appreciable floor area is required for installation, and the screens must be cleaned once or twice daily with high pressure hot water, steam, or degreaser to remove grease buildup. Static wedge wire screens are generally applicable to smaller plants or for industrial installations.

Drum Wedge Wire Screens

For the drum-type wedge wire screen, the screening or straining medium is mounted on a cylinder that rotates in a flow channel. The construction varies, principally with regard to the direction of flow through the screening medium. In the most common arrangements, wastewater flows either into one end (single entry) of the drum and outward through the screen or from both sides (double entry) inward through the screens. In both cases, the screenings collect on the inner surface. As the screen rotates, the debris collected on the inside of the screen is lifted above the operating floor above debris hoppers located inside the screen structure and the debris is washed by a series of washwater jets. The collected debris and washwater then passes to the solids handling systems. In another arrangement, wastewater discharges into the top of the unit and passes through to the interior with solids collection on the exterior. However, this arrangement should only be used on low flowrates.

Traveling Band Screens

Traveling band screens are used for fine screening of influent wastewater at headworks or for secondary effluent ahead of tertiary membrane systems. In the typical application, the water being screened will pass through an opening in the front of the screen and then from the inside of the screen, through screen panels to the outside of the screen totally separating the influent and effluent. As the band of screen panels rotate, the debris collected on the inside of the screen is lifted above deck level by lifting elevators, where it is washed into a debris trough within the head section of the screen. The debris trough will discharge by gravity to the screenings handling equipment. The screens may be controlled either automatically or locally by hand.

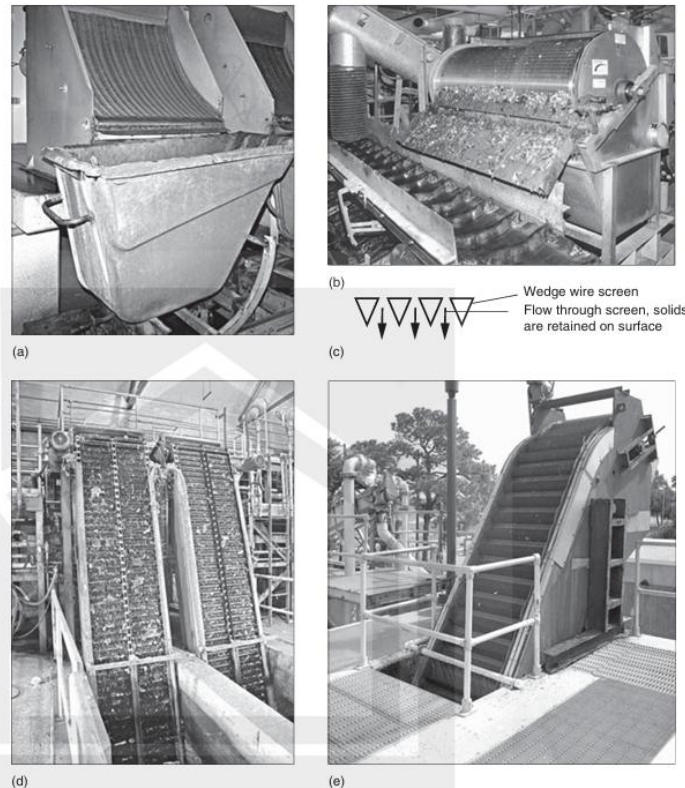
Step Screens

Step screens consist of two step-shaped sets of thin vertical plates, one fixed and one movable. The fixed and movable step plates alternate across the width of an open channel and together form a single screen face. The movable plates rotate in a vertical motion. Through this motion, solids captured on the screen face are automatically lifted up to the next fixed step landing, and are eventually transported to the top of the screen where they are discharged to a collection hopper. The circular pattern of the moving plates provides a self-cleaning feature for each step. Normal ranges of openings between the screen plates are 3 to 6 mm (0.12 to 0.24 in.); however, openings

as small as 1 mm (0.04 in.) are available. Solids trapped on the screen also create a “filter mat” that enhances solids removal performance. In addition to wastewater screening, step screens can be used for removal of solids from septicage, primary sludge, or digested biosolids.

Figure 5-4

Typical fine screens: (a) static wedge wire, (b) wedge-wire drum screen, (c) section through wedge wire screen, (d) traveling band screen, and (e) step screen.



Microscreens

Micro-screening involves the use of variable low-speed (up to 4 rev/min), continuously backwashed, rotating drum screens operating under gravity-flow conditions. The filtering fabrics have openings of 10 to 35 microns (micrometers) and are fitted on the drum periphery. The wastewater enters the open end of the drum and flows outward through the rotating-drum screening cloth. The collected solids are backwashed by high-pressure jets into a trough located within the drum. The principal applications for microscreens are to remove suspended solids from secondary effluent and from stabilization pond effluent.

Typical suspended-solids removal achieved with microscreens ranges from 10 to 80 percent, with an average of 55 percent. Problems encountered with microscreens include incomplete solids removal and inability to handle solids fluctuations. Reducing the rotating speed of the drum and less frequent flushing of the screen have resulted in increased removal efficiencies but reduced capacity.



Mixing

Mixing in water treatment is a vital physical unit operation designed to rapidly disperse treatment chemicals or blend raw water sources. It relies on mechanical or hydraulic energy rather than chemical reactions to achieve uniformity, serving as the essential first step before coagulation, flocculation, and disinfection.

Key Mixing Mechanisms

- **Rapid (Flash) Mixing:** This high-energy physical process uniformly distributes coagulants (like alum or ferric salts) through raw water in less than 30 to 60 seconds. Rapid mixing destabilizes the electrical charges of colloidal particles so they can begin to clump together.
- **Slow Mixing (Flocculation):** A gentle physical agitation process that takes place over 15-45 minutes. It slowly brings the destabilized particles together to form larger, heavier clumps called "flocs" without breaking them apart.

Types of Mixers Used for Rapid Mixing in Wastewater Treatment

1. Static Mixers.

Static inline mixers contain internal vanes or orifice plates that bring about sudden changes in the velocity patterns as well as momentum reversals. Static mixers are principally identified by their lack of moving parts. Typical examples include inline static mixers that contain elements that bring about sudden changes in the velocity patterns as well as momentum reversals (fig: a) and mixers that contain orifice plates and nozzles (fig: b). Static inline mixers are used most commonly for

mixing of chemicals with wastewater. Inline mixers are available in sizes varying from about 12 mm to 3 m 3 3 m open channels. Low-pressure-drop round, square, and rectangular inline static mixers have been developed for chlorine mixing in open channels and tunnels.

2. Inline Mixers

Inline mixers are similar to static mixers but contain a rotating mixing element to enhance the mixing process. Typical examples of inline mixers are illustrated on (c) and (d). In the inline mixer shown on Fig. (c), the power required for mixing is supplied by an external source. For the mixer shown on Fig. (d), the power for mixing is supplied by the energy dissipation caused by the orifice plate and by the power input to the propeller mixer.

3. High-Speed Induction Mixer.

The high-speed induction mixer is an efficient mixing device for a variety of chemicals. A proprietary device, shown on Fig. (e) for chlorine mixing, consists of a motor-driven open propeller that creates a vacuum in the chamber directly above the propeller. The vacuum created by the impeller induces the chemical to be mixed directly from the storage container without the need for dilution water. The high operating speed of the impeller (3450 rev/min) provides a thorough mixing of the chemical that is being added to the water by the high velocity of the fluid leaving the impeller of the mixing device.

4. Pressurized Water Jets.

Pressurized water jet mixers, such as illustrated on Fig. (f), can also be used to mix chemicals. An important design feature of pressurized water jet mixers is that the velocity of the jet containing the chemical to be mixed must be sufficient to achieve mixing in all parts of the pipeline. As shown on Fig. (f), a reactor tube has been added to achieve effective mixing. With pressurized water jet mixers, the power for mixing is provided by an external source (i.e., the solution feed pump).

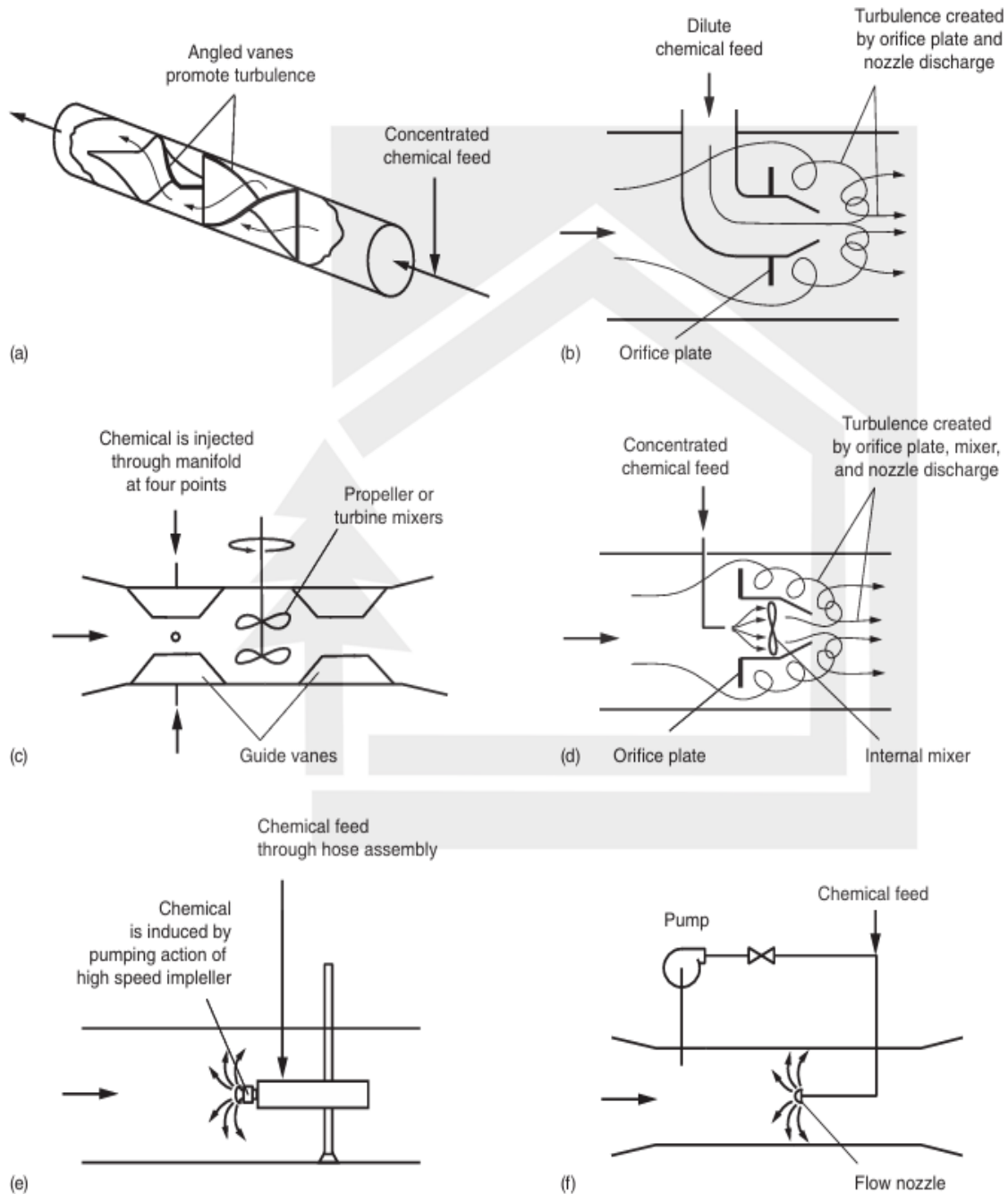


Fig: Typical mixers used in wastewater treatment for rapid mixing: (a) inline static mixer with internal vanes, (b) inline static mixer with orifice used to create internal turbulence for mixing dilute chemicals, (c) inline propeller or turbine mixer, (d) inline mixer with orifice and internal propeller mixer , (e) high speed induction mixer, and (f) pressurized water jet mixer.

Equalization – homogenization – sulphide oxidation

The main aims here are:

- homogenization of the effluent (quantity and quality); and
- sulphide elimination, mostly by catalytic oxidation.

It is very important to keep all particulate matters in suspension, i.e., to avoid settling of solids. This is achieved by using mixing-cum-aeration devices such as diffused-air systems (preferred), Venturi ejectors, and fixed or floating aerators (lately avoided due to lower efficiency and the problem of aerosols). In practice, to play it safe, the volume of the equalization tank corresponds to the total daily effluent discharge.

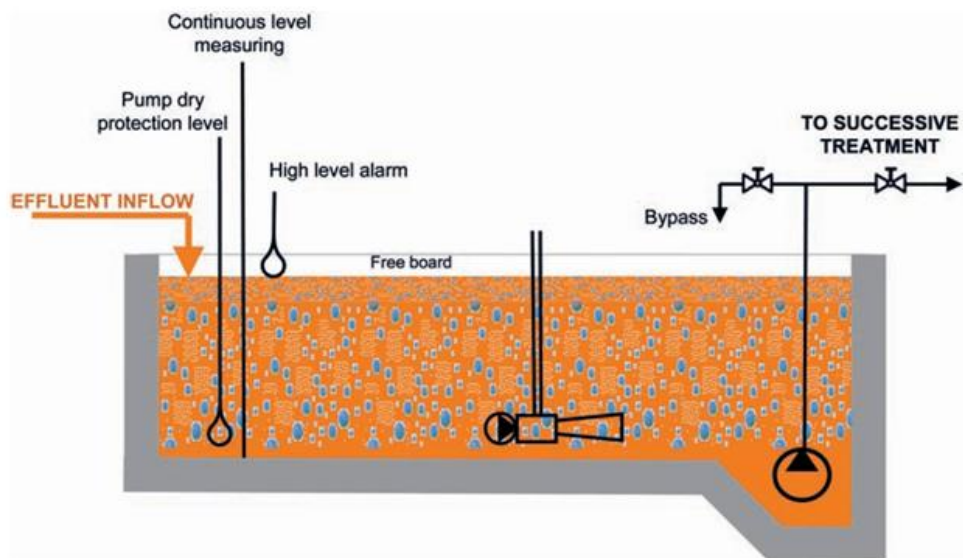


Figure 14. Schematic view of an equalization, homogenization tank

Approximately 1 kg of O_2 is needed to oxidize 1 kg of S^{2-} to thiosulphate, whereas the oxygen transfer efficiency is about 1.5 kg O_2 /kWh (simplified approximation: 1 kg S^{2-} = 1 kg O_2 = 1kWh). Again, in practice, attention is focused on the energy required to keep the solids in suspension (some 50 W/m³), which is then sufficient for sulphide oxidation; the amount of catalyst, $MnSO_4 \cdot 4 H_2O$, industrial purity, is about 20 g per cubic meter of tank capacity.

Whichever the mixing/aeration system chosen, it is necessary to be possible to remove the mixing device without stopping the treatment process. The inlet and the outlet of the equalization tank should be as far away from each other as possible to allow proper mixing (and no short-circuiting).

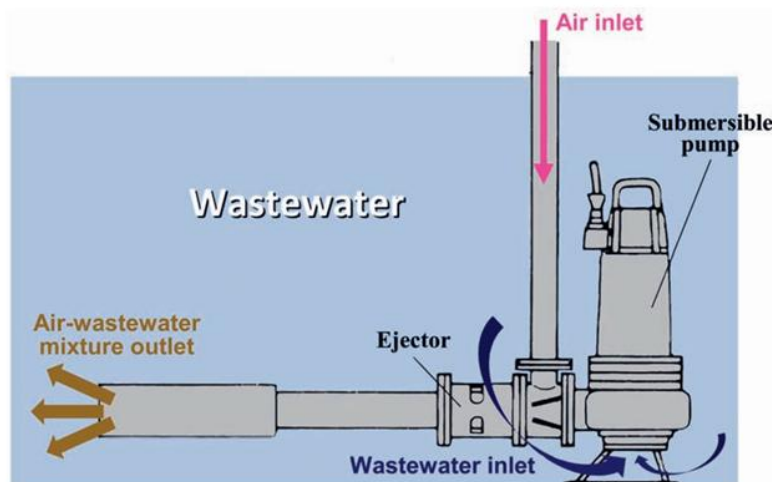


Figure 15. Venturi-type ejector often installed for mixing and aeration in the homogenization tank

A typical equalization tank will have transfer pumps for equalized effluent. The capacity of the pumps is based on tank capacity, transfer time and head. One pump of cast iron with inside parts of stainless steel and one stand-by pump are sufficient unless effluent volume is very high (say 1,500 m³/d or more). The pumping line(s) are also a good place to set an electro-magnetic flow meter.

Sedimentation

The objective of treatment by sedimentation is to remove readily settleable solids and floating material found in wastewater and thus reduce the suspended solids content. Primary sedimentation is typically the first step in the further processing of the wastewater following the removal of coarse solids and grit. Efficiently designed and operated, from 50 to 70 percent of the suspended solids and from 25 to 40 percent of the BOD can be removed by primary sedimentation tanks.

Sedimentation tanks have also been used as stormwater retention tanks to provide a moderate detention period (10 to 30 min) for overflows from either combined sewers or storm sewers. The purpose of sedimentation is to remove a substantial portion of the organic solids that otherwise would be discharged directly to the receiving waters. Sedimentation tanks have also been sized to provide detention periods sufficient for effective disinfection of such overflows.

Sedimentation is a common way in which water is treated. It is a process that helps in removing solids that float and usually settles in the water. The process is carried out with the use of a sedimentation tank that also allows in removing larger solids.

Sedimentation is a treatment used commonly in municipalities. It is basically a physical water treatment process where the use of gravity enables the removal of suspended solids from the water.

The effectiveness of water treatment from sedimentation depends on the size and weight of present particles in the water. Solids that are suspended in water having a similar gravity to that of the water tend to remain suspended, whereas the heavier particles lead to settling. The sedimentation tanks of various shapes and sizes are used in order to carry out the sedimentation process.

Principles of Sedimentation

Sedimentation works on the principle that particles heavier than water will, over time, settle to the bottom of a tank or basin due to gravity. The efficiency of sedimentation depends on several factors:

- a. Particle Characteristics: Size, shape, and density of the particles influence their settling velocity.
- b. Water Temperature: Affects the viscosity of water; warmer temperatures decrease viscosity, allowing particles to settle faster.
- c. Flow Conditions: Laminar flow conditions are ideal for sedimentation, as turbulence can resuspend settled particles.

Types of Sedimentation

1. Discrete Particle Settling (Type I):

- Particles settle individually without interaction.
- Common in the removal of sand and grit in pre-treatment stages.

2. Flocculent Settling (Type II):

- Particles coalesce to form larger flocs, increasing settling rates.
- Occurs after coagulation and flocculation processes.

3. Zone Settling (Type III):

- High concentrations of particles settle as a mass, maintaining a distinct interface with the clearer water above.
- Typical in secondary clarifiers dealing with biological solids.

4. Compression Settling (Type IV):

- Settling occurs under the weight of overlying particles, compressing the sludge layer.
- Relevant in sludge thickening processes.

Sedimentation Tank Design:

Designing an effective sedimentation tank involves optimizing parameters to maximize solids removal efficiency while ensuring operational practicality.

Key Design Parameters:

a.Surface Area: A larger surface area allows for more particles to settle.

b.Detention Time: The time water spends in the tank; sufficient detention time ensures adequate settling.

c.Tank Depth: Adequate depth prevents resuspension of settled solids.

d.Inlet and Outlet Structures: Proper design minimizes turbulence and ensures even flow distribution.

Types of Sedimentation Tanks:

1. Rectangular Tanks:

- Long and narrow, promoting uniform flow.
- Easier to construct in series for large treatment plants.

2. Circular Tanks:

- Radial flow pattern aids in even settling.
- Central sludge collection simplifies removal processes.

3. Inclined Plate or Tube Settlers:

- Incorporate inclined plates or tubes to increase settling surface area within a compact footprint.
- Enhance settling efficiency, allowing for smaller tank sizes.

Operational Considerations

a. Inlet Design: Diffusers or baffles distribute incoming water evenly to reduce turbulence.

b. Outlet Design: Weirs or launders collect clarified water uniformly from the surface.

c. Sludge Removal: Mechanical scrapers or suction systems continuously remove settled sludge to prevent accumulation.

d. Flow Control: Maintaining steady flow rates prevents disturbances that can hinder settling.

Enhancements in Sedimentation

1. Chemical Coagulation and Flocculation:

- Adding coagulants (like alum or ferric salts) neutralizes particle charges, promoting floc formation.
- Flocculants (such as polymers) bind particles into larger aggregates, enhancing settling rates.

2. High-Rate Clarification:

- Ballasted Flocculation: Introduces fine sand or other media as a ballast to increase the weight of flocs.
- Lamella Clarifiers: Use inclined plates to reduce settling distances, increasing efficiency.

3. Dissolved Air Flotation (DAF):

- An alternative to sedimentation for lighter particles.
- Air bubbles attach to particles, causing them to float to the surface for removal.

Primary settling tanks (clarifiers) are either circular (more commonly used) or rectangular with continuous grease (scum) removal at the top and sludge removal at the bottom. The key design parameters are:

- detention time – usually 1 to 2 hours (vertical clarifiers of the Dortmund type);
- surface hydraulic loading, expressed in m^3/m^2 of tank surface per hour or m/h , typically 1 to 2 m^3/m^2 per hour;
- surface solids rate, expressed in kg/m^2 and indicating the quantity of SS crossing the surface area of the tank over a certain time span (hour, day).

The surface solids rate is most frequently used in the design of sludge thickeners but, due to the quantity (4-6 g/l) and flocculent nature of tannery effluent solids, it is useful in controlling the primary sedimentation as well. Circular tanks are generally preferred as recirculation is easier. A mechanical device (scraper) is necessary in larger settling tanks.

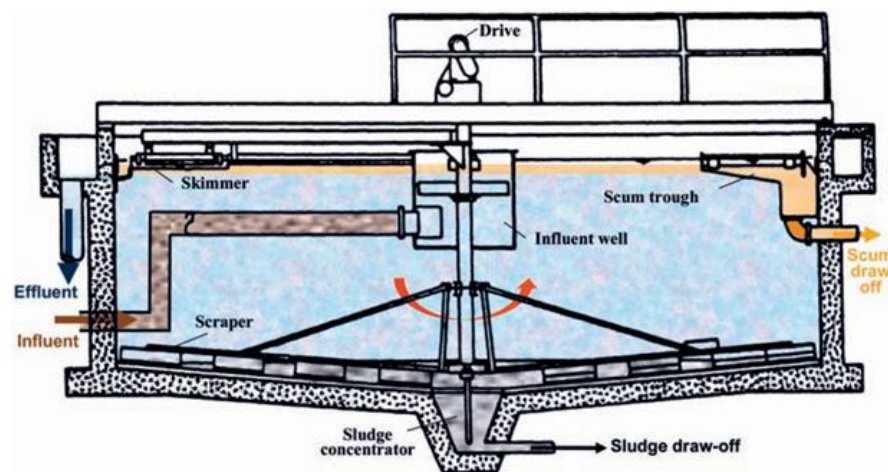


Figure 18. Cross section of a typical circular sedimentation tank



Photo 3. Sedimentation tank (empty)

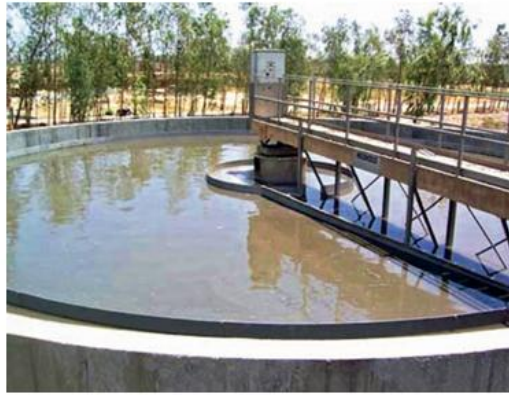


Photo 4. Sedimentation tank in operation

In some cases, mainly due to space shortage, solids are removed by flotation, usually by the dissolved-air flotation (DAF) system.

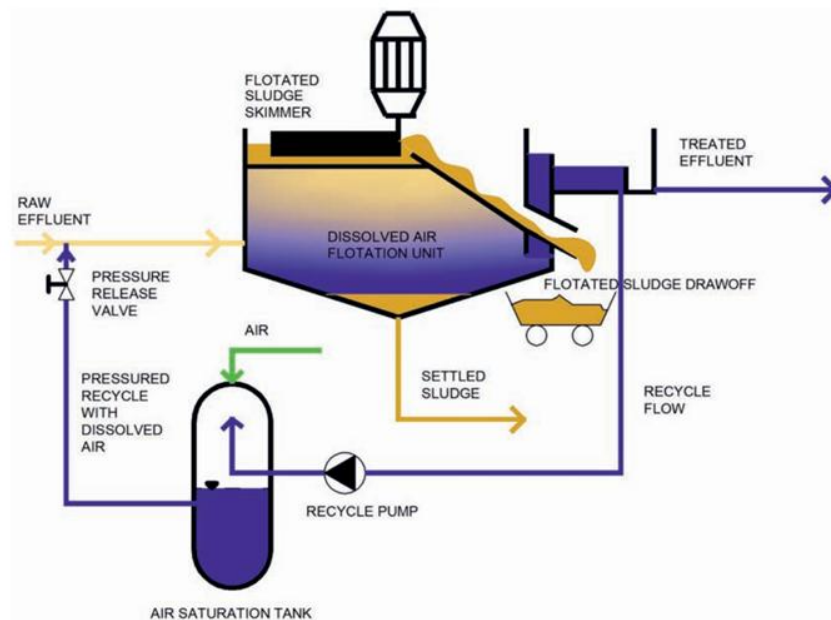


Figure 20. Schematic view of a dissolved-air flotation (DAF) unit

Filtration

Turbidity and algae are removed from raw waters by screens, gravel filters, slow sand, rapid gravity filters or cartridge filters. The difference between slow and rapid sand filtration is not a simple matter of the speed of filtration, but in the underlying concept of the treatment process. Slow sand filtration is essentially a biological process whereas rapid sand filtration is a physical treatment process. Many small private water supplies will rely on cartridge filters consisting of a woven or spun filter within a standard housing.

1. Screens

Screens are effective for the removal of particulate material and debris from raw water and are used on many surface water intakes. Coarse screens will remove weeds and debris while band screens or micro-strainers will remove smaller particles including fish and may be effective in removing large algae. Microstrainers are used as a pre-treatment to reduce solids loading before coagulation or subsequent filtration. A microstrainer may consist of a rotating drum fitted with very fine mesh panels, or may be a fixed mesh that the water flows through. The mesh will ensure that suspended solids, including algae, are retained. The extent of solids removal will be determined by the mesh size and the nature of the raw water. Water will need subsequent treatment downstream and screens should only be considered as a preliminary treatment stage.

2. Gravel filters

Gravel filters may be used to remove turbidity and algae. A larger gravel filter may consist of a rectangular channel or tank divided into several sections and filled with graded gravel (size range 4 to 30mm). The raw water enters through an inlet distribution chamber and flows horizontally through the tank, encountering first the coarse and then the finer gravel. The filtered water is collected in an outlet chamber. Solids removed from the raw water accumulate on the floor of the filter. Gravel filters can operate for several years before cleaning becomes necessary. The size of a gravel filter will depend on water quality, flow rate and size of gravel. A filter can be up to 12m long, 2 to 5m wide and 1 to 1.5m deep. The filter should normally be sized for a flow rate of between 0.5 to 1.0 cubic meters per square meter of filter surface area per hour ($\text{m}^3/\text{m}^2.\text{h}$). A gravel filter will need subsequent treatment downstream, and should only be considered as a preliminary treatment stage.

3. Slow sand filters

Slow sand filters, sometimes preceded by microstrainers or coarse filtration, are used to remove turbidity, algae and microorganisms. Slow sand filtration is a simple and reliable process and is therefore often suitable for the treatment of small supplies provided that sufficient land is available. Slow sand filters usually consist of tanks containing sharp sand (size range 0.15-0.30mm) to a depth of between 0.5 to 1.5m. For small supplies, modular units of 1.25m diameter are available – a tandem installation would occupy a concrete apron of about 8 to 10 m^2 . The raw water flows downwards and turbidity and microorganisms are removed by filtration in the top few centimeters of the sand. A biological layer of sludge, known as the *schmutzdecke*, develops on the surface of the filter that can be effective in removing microorganisms. Treated water is collected in underdrains or pipework at the bottom of the filter. The top few centimetres of sand containing the accumulated solids are removed and replaced periodically. Filter runs of between 2 and 10 weeks are possible, depending on raw water quality and flow rate. Slow sand filters are often operated in tandem; one in service whilst the other is cleaned and time allowed for the *schmutzdecke* to re-establish.

A variant of the slow sand filter, the “Inverness filter”, has been widely used in Scotland. It uses the same grade of sand and operates at the same flow rate as the traditional slow sand filter but the water flows upwards. Filtration is achieved throughout the filter bed and a true schmutzdecke does not develop. The sand is “washed” by opening a valve at the bottom of the filter and allowing the filter bed to drain rapidly.

Slow sand filters should be sized for a water flow rate of between 0.1 and 0.3 m³/m².h. The flow rate should be controlled and the filter designed with a treated water reservoir of sufficient capacity to accommodate fluctuations in demand, and thus permit operation of filters at a steady and continuous rate.

4. Rapid gravity filters

Rapid gravity filters are most commonly used to remove floc from coagulated waters. They may also be used to remove turbidity, algae and iron and manganese from raw waters. Granular activated carbon media may be used to remove organic compounds.

Rapid gravity sand filters usually consist of rectangular tanks containing silica sand and/or anthracite media (size range 0.5 to 1.0 mm) to a depth of between 0.6 and 1.0m. The water flows downwards and solids become concentrated in the upper layers of the bed. Treated water is collected via nozzles in the floor of the filter. The accumulated solids are removed periodically by backwashing with treated water, usually preceded by scouring of the media with air. Frequency of backwashing depends on loading rate and raw water quality and is typically every 24 hours. Backwashing can be initiated automatically after a predetermined headloss has been reached or may be carried out manually. Backwash flowrates will be stipulated by the manufacturer. It is important to achieve the required bed expansion in order to ensure filters are washing properly, but care must be taken to prevent loss of media with too high a flowrate. A dilute sludge that requires disposal is produced which may be discharged to sewer, soak away, or after treatment, to a watercourse provided that any required discharge consent is obtained.

5. Pressure filters

Pressure filters are sometimes used where it is necessary to maintain hydraulic head in order to eliminate the need for additional pumping. The filter bed is enclosed in a cylindrical shell. Small pressure filters capable of treating up to approximately 15m³/h can be manufactured in glass reinforced plastics. Larger pressure filters are manufactured in specially coated steel. Operation and performance are generally as described for the rapid gravity filter and similar facilities are required for backwashing and disposal of the dilute sludge. A similar range of contaminants can be removed depending on the filter medium.

Principle of chemical treatment

The principle of chemical treatment of water is to improve water quality by adding chemicals that remove, neutralize, or destroy impurities and harmful substances. Chemical treatment is commonly used in drinking water purification, wastewater treatment, and industrial water processing.

The chemical treatment of water relies on applying specific chemical agents such as coagulants, disinfectants, and oxidants—to neutralize, aggregate, or destroy impurities. Some of chemical treatment process are described as below:

Coagulation

Coagulation in water treatment is a process used to remove suspended particles, colloids, and other impurities from water. This process involves adding specific chemicals, known as coagulants, to raw water. The primary goal is to destabilize the fine particles that are too small to be removed by gravity alone. Once destabilized, these particles clump together into larger aggregates called flocs, which can then be removed more easily through sedimentation or filtration.

The core principle of coagulation lies in altering the physical and chemical characteristics of particles in the water. Most particles suspended in water carry a negative electrical charge, which keeps them from clumping together. Coagulants, which are typically positively charged chemicals, neutralize these negative charges. This neutralization reduces the repulsive forces between particles, allowing them to aggregate into larger, floc-like structures.

How does coagulation treatment work?

1. Addition of Coagulants

The coagulation process begins with the addition of coagulants to raw water. This step is crucial for initiating the transformation of suspended particles into larger, removable aggregates. Here's how it typically works:

- a) Coagulant Dosing: Coagulants, such as aluminum sulfate or ferric chloride, are carefully dosed into the raw water. The amount of coagulant added is determined based on factors such as water quality, particle concentration, and the specific type of coagulant being used.
- b) Mixing: After dosing, the water is rapidly mixed in a process called rapid mixing or flash mixing. This ensures that the coagulant is evenly distributed throughout the water and reacts uniformly with the suspended particles. Effective mixing is essential for the coagulant to interact properly with the particles and initiate the coagulation process.

2. Formation of Flocs

Once coagulants are introduced and mixed, the next step is the formation of flocs. This is where the fine particles in the water start to aggregate into larger clumps. Here's how the floc formation process unfolds:

- a) Charge Neutralization: Coagulants neutralize the negative electrical charges on the suspended particles, which reduces their repulsion and allows them to attract each other.
- b) Particle Agglomeration: As particles begin to lose their repulsive charges, they start to stick together. The neutralized particles bond with one another, forming larger clusters known as flocs.
- c) Floc Growth: The flocs continue to grow as more particles join them. This aggregation process is facilitated by the coagulant, which binds the particles together into larger, more visible floc formations.

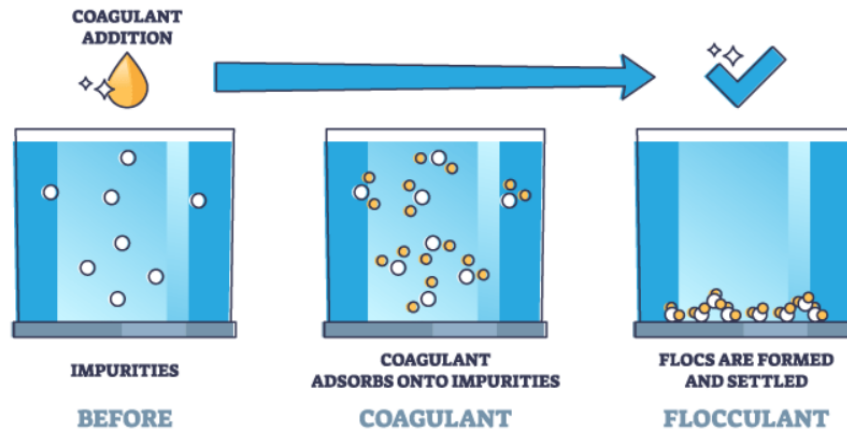
3. Sedimentation and Removal

The final step in the coagulation process involves the removal of the flocs from the water. This is achieved through sedimentation and, in some cases, additional filtration. Here's how these steps work:

- a) Sedimentation: After floc formation, the water is allowed to settle in a sedimentation basin or clarifier. In this basin, the flocs, now heavier than the surrounding water, begin to settle to the bottom due to gravity. This process is known as sedimentation or settling. The settled flocs form a sludge layer at the bottom of the basin.
- b) Removal of Settled Flocs: The clear water above the settled sludge is then typically withdrawn for further treatment or distribution. The sludge containing the accumulated flocs is removed periodically from the bottom of the basin.
- c) Additional Filtration (if required): In some water treatment plants, additional filtration may follow sedimentation to further polish the water. This step helps to ensure that any remaining fine particles or residual flocs are removed before the water undergoes final disinfection.

By effectively managing each step of the coagulation process (coagulant addition, floc formation, and sedimentation) water treatment facilities can significantly improve water quality, ensuring that it meets safety and clarity standards before distribution.

WATER COAGULATION



Types of Coagulants

A. Chemical Coagulants

Chemical coagulants are widely used in water treatment for their effectiveness in promoting coagulation and flocculation. Here are two of the most common chemical coagulants:

1. Aluminum Sulfate (Alum)

How It Works:

Alum reacts with water to form aluminum hydroxide, which acts as a floc-forming agent. When added to water, alum neutralizes the negative charges on suspended particles, causing them to aggregate into larger flocs.

Advantages:

- **Effectiveness:** Alum is highly effective in a wide range of water conditions and pH levels.
- **Cost-Effective:** It is relatively inexpensive and widely available.

Disadvantages:

- **Residual Sludge:** The use of alum produces a significant amount of residual sludge, which requires proper disposal.
- **pH Adjustment:** It can lower the pH of the water, potentially requiring additional pH adjustment.

2. Ferric Chloride

How It Works:

Ferric chloride reacts with water to form ferric hydroxide, which helps in the formation of flocs. It is effective in neutralizing charges on particles and facilitating their aggregation.

Advantages:

- **Versatility:** Ferric chloride works effectively across a range of pH levels and water conditions.
- **Reduced Sludge Volume:** It generally produces less sludge compared to alum.

Disadvantages:

- **Higher Cost:** It can be more expensive than alum.
- **Corrosive Nature:** Ferric chloride is more corrosive, which can lead to increased maintenance costs for equipment.

B. Natural Coagulants

Natural coagulants are derived from natural sources and offer an eco-friendly alternative to traditional chemical coagulants. Here are some notable examples:

1. Moringa Oleifera Seeds

How They Work:

Moringa seeds contain natural coagulant proteins that can effectively destabilize particles in water. When crushed and added to water, these proteins help to aggregate suspended particles into flocs.

Advantages:

- **Eco-Friendly:** Moringa seeds are a renewable resource and are biodegradable, minimizing environmental impact.
- **Low Cost:** They can be a low-cost alternative in regions where Moringa is readily available.
- **Health Benefits:** Moringa is rich in nutrients and has additional health benefits beyond water treatment.

Disadvantages:

- **Effectiveness Variability:** The effectiveness can vary based on the type of water and the preparation method.
- **Limited Use:** It may not be as effective as chemical coagulants in certain conditions or at higher scales.

2. Chitosan

How It Works:

Chitosan is derived from chitin, a biopolymer found in crustacean shells. It acts as a flocculant by neutralizing charges and binding particles.

Advantages:

- **Biodegradable:** It is environmentally friendly and breaks down naturally.
- **High Efficiency:** Chitosan is effective in a wide range of water conditions.

Disadvantages:

- **Higher Cost:** Chitosan can be more expensive than traditional chemical coagulants.
- **Availability:** It is less commonly available compared to conventional coagulants.

Importance of Coagulation in Water Treatment

1. Improving Water Clarity

Coagulation plays a pivotal role in enhancing water clarity, which is a fundamental aspect of water quality. The process involves adding coagulants to raw water, which destabilizes fine particles and aggregates them into larger clusters known as flocs. These flocs are significantly larger and denser than the original particles, making them more susceptible to removal through sedimentation and filtration. As a result:

- a) **Visible Clarity:** The removal of suspended particles through coagulation greatly improves the visual clarity of water, making it more aesthetically pleasing and acceptable for consumption and recreational use.
- b) **Reduced Turbidity:** Coagulation effectively reduces water turbidity by aggregating and removing particles that contribute to cloudiness, thereby making the water more transparent.

2. Contaminant Removal

Coagulation is crucial for removing various contaminants, including pathogens and organic matter, from water. Here's how coagulation contributes to contaminant removal:

- a) **Pathogen Removal:** Many pathogens, such as bacteria and viruses, are often associated with suspended particles. By removing these particles through coagulation, the concentration of pathogens in the water is significantly reduced, thereby enhancing water safety and reducing the risk of waterborne diseases.

- b) Organic Matter Removal: Coagulation helps in the removal of organic compounds, including natural organic matter (NOM) and other pollutants. This is particularly important in preventing the formation of disinfection byproducts during subsequent water treatment stages.

3. Load Reduction

Coagulation impacts subsequent filtration and disinfection processes by reducing the overall load of suspended particles and contaminants. Here's how it benefits the overall water treatment process:

- a) Eased Filtration: By aggregating fine particles into larger flocs, coagulation reduces the particle load on filtration systems. This makes the filtration process more efficient by preventing clogging and reducing the frequency of filter maintenance.
- b) Enhanced Disinfection: Coagulation lowers the concentration of particulate matter and organic compounds, which can otherwise react with disinfectants to form harmful byproducts. This helps in ensuring that disinfection processes, such as chlorination or UV treatment, are more effective in eliminating pathogens and improving water safety.
- c) Operational Efficiency: The reduction of particle load in the water stream enhances the efficiency of the entire water treatment process, leading to lower operational costs and improved water quality.

In summary, coagulation is a critical step in water treatment that improves water clarity, aids in contaminant removal, and reduces the burden on subsequent filtration and disinfection processes. By optimizing these processes, coagulation ensures that the water is not only visually appealing but also safe for consumption and other uses.

Coagulation Control

Monitoring and Adjustment

Effective coagulation control is essential for optimizing the water treatment process and ensuring that water quality meets safety standards. Key parameters that need to be carefully monitored and adjusted include:

- Coagulant Dosage: The amount of coagulant added to the water must be precise. Too little coagulant can result in inadequate particle aggregation, while too much can lead to excessive sludge production and higher costs. Regular monitoring helps in maintaining the right dosage for effective coagulation.
- pH Levels: The pH of the water significantly impacts the coagulation process. Coagulants operate best within specific pH ranges. For example, alum is most effective in slightly

acidic to neutral pH levels. Monitoring and adjusting the pH ensures optimal coagulant performance and floc formation.

- **Temperature:** Water temperature can affect the rate of coagulation and floc formation. Colder temperatures can slow down these processes, while higher temperatures might increase the rate of reactions. Adjustments in coagulant dosage or process parameters might be necessary to account for temperature variations.

Flocculation

Flocculation is a key process in water treatment where fine particles suspended in water are brought together into larger clumps called flocs. These flocs are easier to separate from the water, making the treatment process more effective. The formation of flocs is achieved by adding specific chemicals known as **flocculants**, which help particles bind together.

Flocculation occurs when flocculants are introduced into the water. These flocculants work by neutralizing the electrical charges that usually keep particles apart, allowing them to come together and form larger aggregates. Once these particles are grouped into flocs, they can be more easily removed from the water through processes like sedimentation or filtration. Common flocculants include aluminum sulfate (alum), iron salts, and organic polymers, each chosen based on the specific characteristics of the water being treated and the nature of the contaminants present.

Relation to Coagulation

These two processes are intimately connected and often work sequentially to achieve optimal water purification:

Coagulation:

This is the initial step where coagulants, such as aluminum sulfate or iron salts, are added to water to destabilize and neutralize the charges of suspended particles. This neutralization allows particles to come closer together but doesn't necessarily form large aggregates.

Flocculation:

Following coagulation, flocculation involves the gentle mixing of water to encourage the neutralized particles to collide and stick together, forming larger and settleable flocs. Additional flocculants may be added during this stage to facilitate and strengthen floc formation.

The combination of coagulation and flocculation ensures efficient removal of a wide range of contaminants by transforming dispersed and stable particles into large aggregates that can be easily separated from the water. This sequential process is fundamental in both drinking water and wastewater treatment, leading to safer and cleaner water for various uses.

Precipitation

Chemical precipitation is a critical process in the treatment of wastewater. This process involves the addition of chemicals to wastewater to facilitate the removal of pollutants. The pollutants, once combined with the chemicals, form precipitates that can be easily separated from the water. This method is widely used in wastewater treatment plants across the globe due to its efficiency and cost-effectiveness.

The process of chemical precipitation is a complex one, involving various stages and a range of chemicals. It is a process that requires a deep understanding of chemistry and environmental science. This article aims to provide a comprehensive understanding of the process, its applications, and its importance in the field of wastewater treatment.

Process of Chemical Precipitation

Chemical precipitation is a physical-chemical process that involves the transformation of dissolved substances into solid particles. This transformation is achieved by altering the physical and chemical conditions of the solution. The process begins with the addition of a precipitating agent, which reacts with the dissolved substances to form insoluble precipitates.

The precipitates formed are larger and heavier than the dissolved substances, allowing them to settle at the bottom of the treatment tank. The settled precipitates, known as sludge, are then separated from the treated water. The treated water can then be discharged into the environment or further treated, depending on the level of treatment required.

Chemicals Used in the Process

The choice of chemicals used in the process of chemical precipitation is crucial. The chemicals must be able to react with the pollutants in the wastewater to form insoluble precipitates. Commonly used chemicals include aluminum sulfate, ferric chloride, and lime. The choice of chemical depends on the nature of the pollutants and the specific requirements of the treatment process.

Aluminum sulfate, also known as alum, is commonly used for the removal of phosphates and other suspended solids. Ferric chloride is effective in removing heavy metals and other inorganic pollutants. Lime, on the other hand, is used for the removal of hardness and the adjustment of pH. The use of these chemicals is governed by various regulations to ensure the safety and effectiveness of the treatment process.

Stages of the Process

The process of chemical precipitation involves several stages. The first stage is the addition of the precipitating agent. This is followed by rapid mixing to ensure the uniform distribution of the

chemical. The next stage is flocculation, where gentle mixing allows the formation of larger flocs from the precipitates.

The flocs then settle at the bottom of the tank in the sedimentation stage. The settled sludge is then removed and treated separately. The treated water is then discharged or further treated. Each stage of the process is carefully controlled to ensure the effective removal of pollutants

PROCESS TYPES

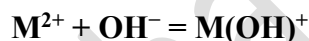
The chemical equilibrium relationship in precipitation that affects the solubility of the component(s) can be achieved by a variety of means. One or a combination of the following processes induces the precipitation reactions in a water environment.

1. Hydroxide Precipitation

Dissolved heavy metal ions can be chemically precipitated as hydroxide for removal by physical means such as sedimentation or filtration. The process uses an alkaline agent to raise the pH of the water that causes the solubility of metal ions to decrease and thus precipitate out of the solvent. The optimum pH at which metallic hydroxides are least soluble varies with the type of metal ion as shown in **Fig 1**. A simple form of the hydroxide precipitation reaction may be written as:



The product formed is an insoluble metal hydroxide. If the pH is below the optimum of precipitation, a soluble metal complex will form:



Hydroxide precipitation is also affected by the presence of organic radicals that can form chelates and mask the typical precipitation reaction:



Reagents commonly used to affect the hydroxide precipitation include alkaline compounds such as lime or caustic soda (sodium hydroxide). Lime in the form of quicklime or un-slaked lime, CaO, and hydrated lime, Ca(OH)₂, can be used. Lime is generally made into wet suspensions or slurries before introduction into the treatment system. The precise steps involved in converting lime from the dry to the wet stage will vary according to the size of the operation and the type and form of lime used. In the smallest plants, bagged hydrated lime is often charged manually into a batch-mixing tank with the resulting “milk of lime” (or slurry) being fed by means of a solution feeder to the treatment process.

Where bulk hydrate lime is used, some type of dry feeder charges the lime continuously to either a batch or continuous mixer. A solution feeder transfers lime to the point of application.

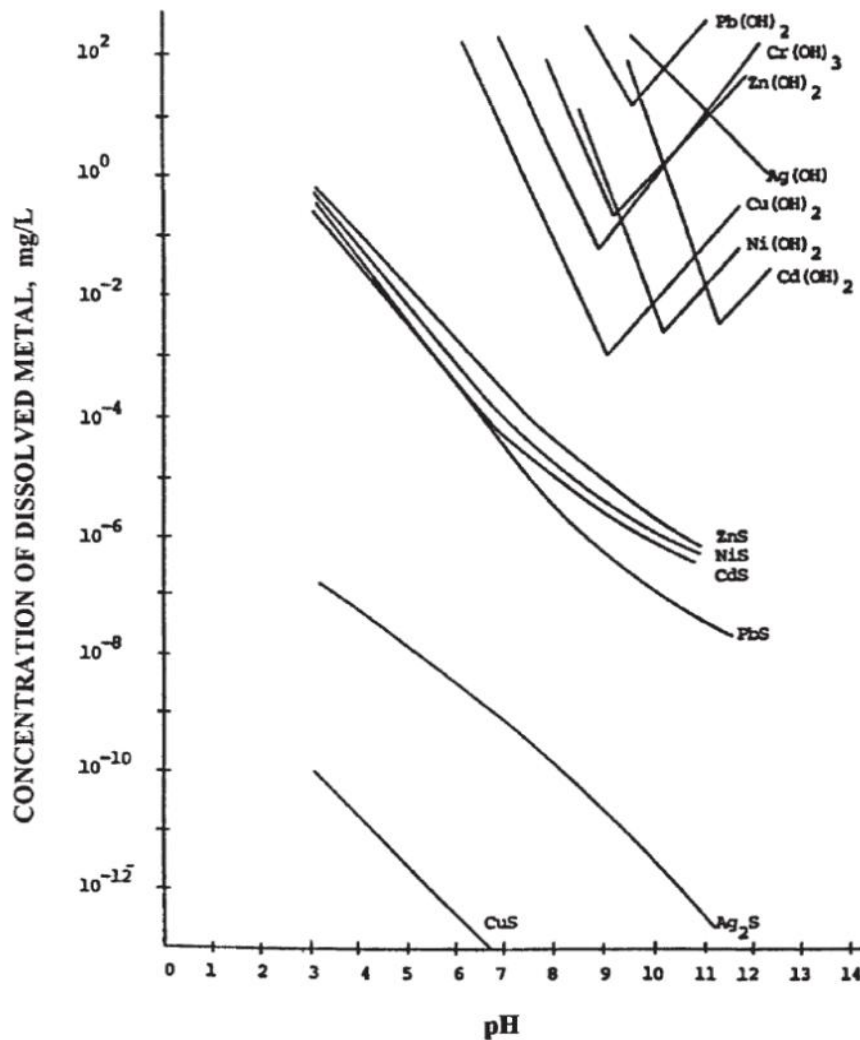


Fig. 1. Solubility of metal hydroxides and sulfides as a function of pH.

With bulk quicklime, a dry feeder is also used to charge a slaking device, where the oxides are converted to hydroxides, producing a paste or slurry. The slurry is then further diluted to milk-of-lime before being fed by gravity or pumping into the process. Dry feeders can be of the volumetric or gravimetric type. Caustic soda, in the form of 6–20% aqueous solution, is fed directly to the treatment system and does not require any dispensing and mixing equipment. The treatment chemicals may be added to a flash mixer or rapid-mix tank, or directly to the sedimentation device. Because metal hydroxides tend to be colloidal in nature, coagulation agents may also be added to facilitate settling.

2. Sulfide Precipitation

Both “soluble” sulfides such as hydrogen sulfide or sodium sulfide and “insoluble” sulfides such as ferrous sulfide may be used to precipitate heavy metal ions as insoluble metal sulfides. Sodium sulfide and sodium bisulfide are the two chemicals commonly used, with the choice between these two precipitation agents being strictly an economic one. Metal sulfides have lower solubilities than hydroxides in the alkaline pH range and also tend to have low solubilities at or below the neutral pH value (**Fig. 1**).

The basic principle of sulfide treatment technology is similar to that of hydroxide precipitation. Sulfide is added to precipitate the metals as metal sulfides and the sludge formed is separated from solution by gravity settling or filtration. Several steps enter into the process of sulfide precipitation:

1. Preparation of sodium sulfide. Although there is often an abundant supply of this product from by-product sources, it can also be made by reduction of sodium sulfate. The process involves an energy loss in the partial oxidation of carbon (such as that contained in coal) as follows:



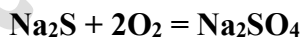
Sodium sulfate + carbon = metallic sulfide + carbon dioxide

2. Precipitation of the pollutant metal (M) in the waste stream by an excess of sodium sulfide



Sodium sulfide + metallic sulfate = metallic sulfide + sodium sulfate

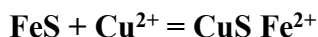
3. Physical separation of the metal sulfide in thickeners or clarifiers, with reducing conditions maintained by excess sulfide ion.
4. Oxidation of excess sulfide by aeration:



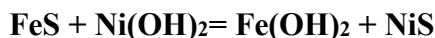
Sodium sulfide + oxygen = sodium sulfate

Because of the toxicity of both the sulfide ion and hydrogen sulfide gas, the use of sulfide precipitation may require both pre- and posttreatment and close control of reagent additions. Pretreatment involves raising the pH of water to between 7 and 8 to reduce the formation of obnoxious hydrogen sulfide gas. The pH adjustment may be accomplished at essentially the same point as the sulfide treatment, or by addition of a solution containing both sodium sulfide and a strong base (such as caustic soda). The posttreatment consists of oxidation by aeration or chemical oxidation to remove excess sulfide, a toxic substance.

A recently developed and patented process to eliminate the potential hazard of excess sulfide in the effluent and the formation of gaseous hydrogen sulfide uses ferrous sulfide as the sulfide source. The fresh ferrous sulfide is prepared by adding sodium sulfide to ferrous sulfate. The ferrous sulfide slurry formed is added to water to supply sufficient sulfide ions to precipitate metal sulfides, which have lower solubilities than ferrous sulfide. Typical reactions are:



Ferrous sulfide + copper ion = insoluble copper sulfide + iron ion



Ferrous sulfide + nickel hydroxide = ferrous hydroxide + insoluble nickel sulfide.

A detention time of 10–15 min is sufficient to allow the reaction to go to completion.

Ferrous sulfide itself is also a relatively insoluble compound. Thus, the sulfide ion concentration is limited by the solubility of ferrous sulfide, which amounts to about 0.02 mg/L, and the inherent problems associated with conventional sulfide precipitation are minimized.

3. Cyanide Precipitation

Cyanide precipitation, although a method for treating cyanide in wastewater, does not destroy the cyanide molecule, which is retained in the sludge that is formed. Reports indicate that during exposure to sunlight, the cyanide complexes can break down and form free cyanide. For this reason the sludge from this treatment method must be disposed of carefully. Cyanide may be precipitated and settled out of wastewater by the addition of zinc sulfate or ferrous sulfate, which forms zinc ferrocyanide or ferro- and ferri-cyanide complexes. In the presence of iron, cyanide will form extremely stable cyanide complexes.

4. Carbonate Precipitation

Carbonate precipitation may be used to remove metals either by direct precipitation using a carbonate reagent such as calcium carbonate or by converting hydroxides into carbonates using carbon dioxide. The solubility of most metal carbonates is intermediate between hydroxide and sulfide solubilities; in addition, carbonates form easily filtered precipitates.

5. Co-precipitation

In co-precipitation, materials that cannot be removed from solution effectively by direct precipitation are removed by incorporating them into particles of another precipitate, which is separated by settling, filtration, or flotation.

Advantages of Chemical Precipitation

The primary advantage of chemical precipitation is its simplicity. The process involves the addition of a precipitating agent, followed by mixing and sedimentation. This simplicity makes the process easy to implement and operate. The process is also cost-effective, requiring relatively low capital and operational costs.

Another advantage of chemical precipitation is its versatility. The process can be used for the removal of a wide range of pollutants, including heavy metals, phosphates, and suspended solids. This versatility makes the process suitable for a wide range of wastewater treatment applications. Furthermore, the process is scalable, making it suitable for both small and large-scale treatment plants.

Disadvantages of Chemical Precipitation

The main disadvantage of chemical precipitation is the production of sludge. The precipitates formed during the process settle at the bottom of the tank, forming a sludge that requires further treatment. The treatment of this sludge can be challenging and costly, adding to the overall cost of the treatment process.

Another disadvantage of chemical precipitation is its sensitivity to changes in the characteristics of the wastewater. Changes in the pH, temperature, or concentration of pollutants can affect the effectiveness of the process. This requires careful monitoring and control of the process conditions. Despite these challenges, chemical precipitation remains a vital tool in the treatment of wastewater.

Ion Exchange Process

The ion exchange process is a chemical technique used to replace undesirable ions in water with more desirable ones through the use of synthetic resins. These resins are specially designed to attract and hold ions from water while releasing their ions in return.

The ion exchange process in water treatment is highly versatile and is commonly applied to soften water, remove contaminants, and purify water for specialized industrial uses.

The Principle of Ion Exchange

The ion exchange process operates on a simple principle: ions are exchanged between a liquid (water) and a solid (resin) based on their charge. Resins are classified into two main types:

1. Cation Exchange Resins

- Target positively charged ions (cations) such as calcium (Ca^{2+}), magnesium (Mg^{2+}), and sodium (Na^+).
- These resins replace unwanted cations in water with hydrogen (H^+) or sodium ions.

- Example: Softening hard water by exchanging calcium and magnesium ions with sodium ions.

2. Anion Exchange Resins

- Target negatively charged ions (anions) such as chloride (Cl^-), sulfate (SO_4^{2-}), and nitrate (NO_3^-).
- These resins replace undesirable anions with hydroxide (OH^-) or other beneficial ions.
- Example: Removing nitrates or sulfates from drinking water.

Cation exchange

Cation exchange is used for both water softening and deionization of cation contaminants. Cation resins are designed to attract cations, which are contaminants having a positive charge. The resin starts with hydrogen ions (H^+) on the surface. When water passes through the resin, cations will attract and adhere to the resin surface, thereby releasing (exchanging) the hydrogen ions. See Figure 2.

The treatment process is always occurring as water flows continuously through the IX vessel. However, once the resin is essentially full of cations, a breakthrough of contaminants starts to occur. At this point, the resin should be regenerated by soaking it in acid, such as sulphuric or hydrochloric acid. The acid has the chemical drive to strip off the cations from the resin and exchange back the hydrogen ions. It is common to have multiple vessels in parallel to allow for one vessel to be out of service during regeneration while the remaining vessels continue treatment.

Common cationic contaminants removed by IX include:

- Calcium (Ca^{2+})
- Iron (Fe^{3+})
- Magnesium (Mg^{2+})
- Manganese (Mn^{2+})
- Potassium (K^+)
- Sodium (Na^+)

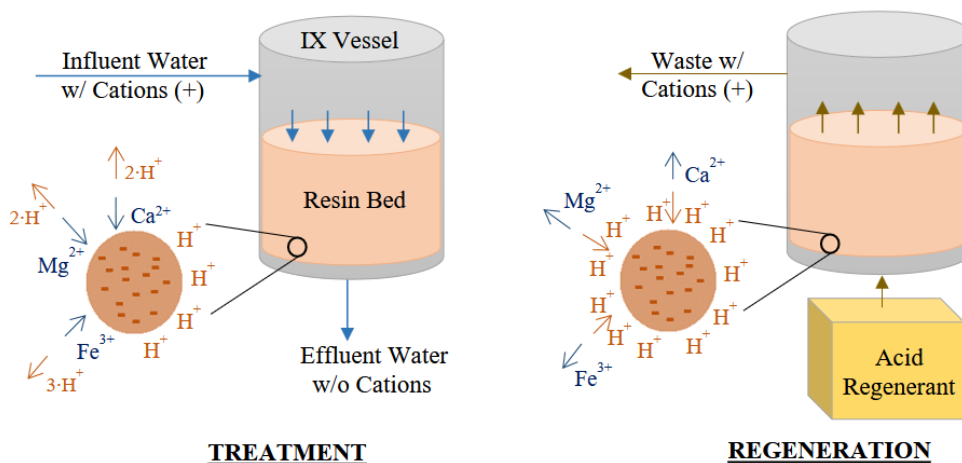


Figure 2: Schematic of a cationic fixed-bed IX system during treatment (left) and regeneration (right). The cation resin beads are negatively charged and attract positive ions such as calcium, magnesium, and iron, which are exchanged with hydrogen ions (H^+). During regeneration, the hydrogen ions are restored to the resin surfaces.

1. Salt-based Water Softeners

Salt-based water softeners target the removal of calcium and magnesium to soften the water. Instead of hydrogen ions, sodium or potassium ions cover the resin and are used for exchange. This allows regeneration with a brine solution ($\text{NaCl} + \text{H}_2\text{O}$), which is lower cost and safer to handle than an acid solution. See Figure 3 for typical arrangements.

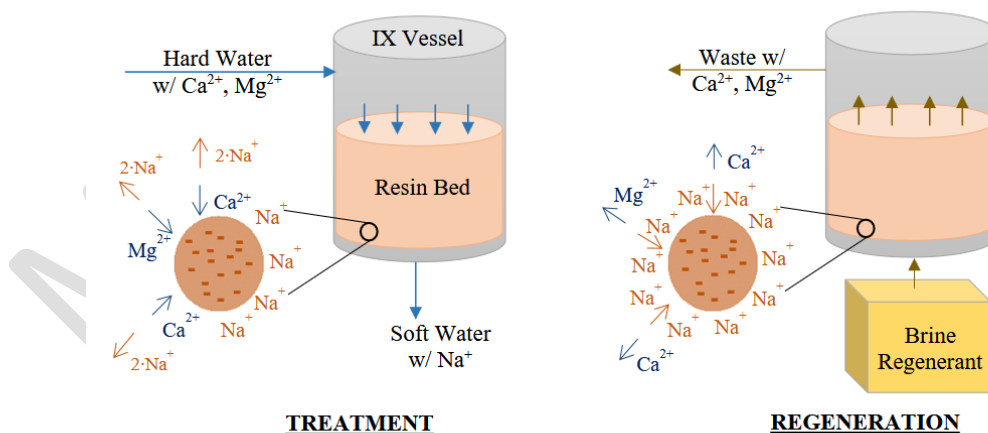


Figure 3: Schematic of a salt-based IX system during treatment (left) and regeneration (right). The cation resin beads are negatively charged and attract calcium and magnesium, which are exchanged with sodium ions (Na^+). During regeneration, a brine solution restores the sodium ions to the resin surfaces.

2. Cation Resins

Resin facilitates the exchange of ions. Resin is a collection of small insoluble beads, nodules, or granules. The most common resin is small organic polymer beads (0.25 to 1.43 mm radius) that are yellow or orange. The beads can be porous to provide a greater surface area. See Figure 5 for examples.



Figure 5: Examples of resin beads.

Cation resins may be either strongly or weakly acidic:

- Strongly acidic: Commonly include sulfonic acid groups, such as sodium polystyrene sulfonate or polyAMPS. Most commercial resins are made of polystyrene sulfonate.
- Weakly acidic: Commonly include carboxylic acid groups.

The beads are manufactured so the surfaces have weak bonds with hydrogen or sodium ions. During treatment, these bonds release and stronger bonds form with cation contaminants such as calcium, magnesium, and iron. See Figure 6 for a chemical depiction of a bead surface.

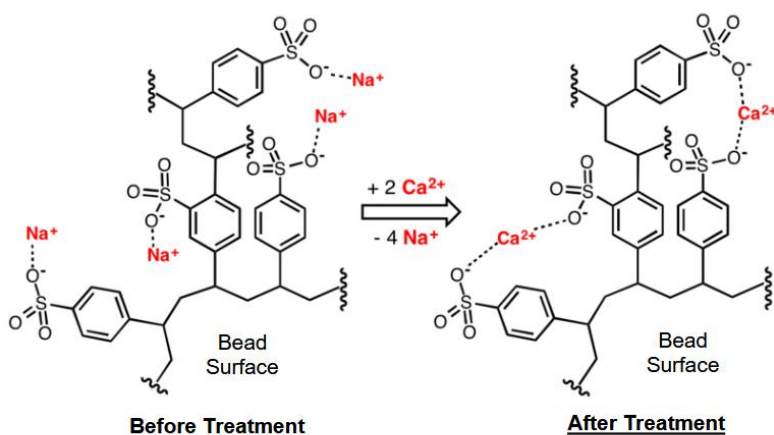


Figure 6: Skeletal structure of a cation bead surface, before treatment (left) with sodium ions on the surface, and after treatment (right) with calcium ions on the surface.

For cation resin, the following is the order of preference, from greatest to least:

- Barium (Ba^{2+})
- Lead (Pb^{2+})
- Mercury (Hg^{2+})
- Calcium (Ca^{2+})
- Nickel (Ni^{2+})
- Cadmium (Cd^{2+})
- Copper (Cu^{+})
- Zinc (Zn^{2+})
- Iron (Fe^{3+})
- Magnesium (Mg^{2+})
- Potassium (K^{+})
- Ammonia (NH_4^{+})
- Sodium (Na^{+})
- Hydrogen (H^{+})

3. Chelating Resins

Chelating resins are cation resins that are manufactured to have a higher-than-normal selectivity towards removing heavy metal cations, including the following:

- Barium (Ba^{2+})
- Cadmium (Cd^{2+})
- Cobalt (Co^{2+})
- Copper (Cu^{+})
- Chromium (Cr^{3+})
- Lead (Pb^{2+})
- Manganese (Mn^{2+})
- Nickel (Ni^{2+})
- Zinc (Zn^{2+})

With a normal cation resin, calcium will be bonded before most heavy metals, which results in frequent regeneration and unwanted pass-through of heavy metals. With a chelating resin, the heavy metals will bond before calcium.

Chelating resins come in the following forms:

1. Sodium form: Starts with sodium ions on the resin surface and is regenerated with brine. This is the most common form and has the greatest selectivity for heavy metals.
2. Hydrogen form: Starts with hydrogen ions on the resin surface and is regenerated with acid. Regeneration is very effective because heavy metals easily release at low pH when the acid solution is introduced.

A drawback to chelating resins is the cost is approximately double of other cation resins.

Anion Exchange

Anion exchange is used to remove a variety of anion contaminants. Anion resins are designed to attract anions, which are contaminants having a negative charge. The resin starts with hydroxide (OH^-) on the surface. When water passes through the resin, anions will attract and adhere to the resin surface, thereby releasing (exchanging) the hydroxide ions. See Figure 7.

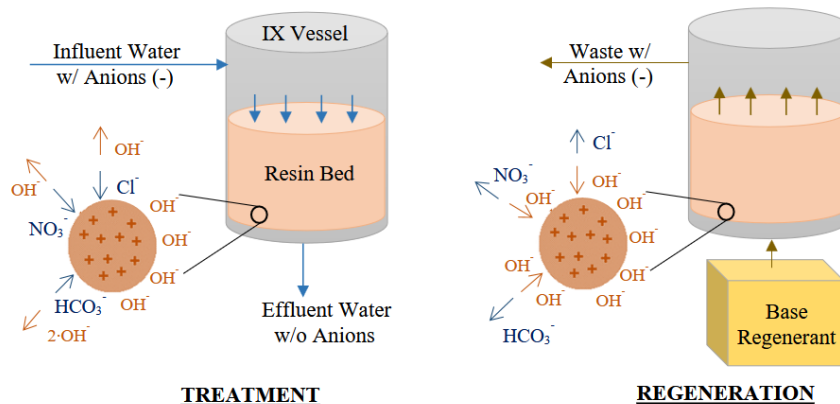


Figure 7: Schematic of an anionic fixed-bed IX system during treatment (left) and regeneration (right). The anion resin beads are positively charged and attract negative ions such as chlorides, nitrates, and bicarbonate, which are exchanged with hydroxide ions (OH^-). During regeneration, the hydroxide ions are restored to the resin surfaces.

The resin should be regenerated by soaking it in a strongly basic solution, such as sodium hydroxide. The base has the chemical drive to strip off the anions from the resin and exchange back the hydroxide ions.

Common anionic contaminants removed by IX include:

- Alkalinity/Carbonate (CO_3^{2-} , HCO_3^-)

- Chloride (Cl^-)
- Nitrate (NO_3^-)
- Sulfate (SO_4^{2-})
- Silica (SiO_2)
- Color
- Total Organic Carbon (TOC)
- Per- and Polyfluorinated Substances (PFAS)

Anion Resins

Anion resins are similar in size and texture to cation resins.

Anion resins may be either strongly or weakly basic:

- Strongly basic: Commonly include quaternary amino groups, for example, trimethylammonium groups, such as polyAPTAC.
- Weakly basic: Commonly include primary, secondary, and/or tertiary amino groups, such as polyethylene amine.

The beads are manufactured so the surfaces have weak bonds with hydroxide ions. During treatment, these bonds release and stronger bonds form with anion contaminants such as chlorides, nitrates, and sulfates.

For anion resins, the following is the order of preference, from greatest to least:

- Perchlorate (ClO_4^-)
- Iodide (I^-)
- Sulfate (SO_4^{2-})
- Nitrate (NO_3^-)
- Bisulfite (HSO_3^-)
- Bromine (Br^-)
- Chloride (Cl^-)
- Cyanide (CN^-)
- Bicarbonate (HCO_3^-)
- PFAS (various)
- Acetate (CH_3COO^-)

- Hydroxide (OH^-)
- Fluoride (F^-)

How the Ion Exchange Process in Water Treatment Works?

1. Pretreatment

- Before the ion exchange process begins, water is pretreated to remove suspended solids, organic matter, and other impurities that could damage or clog the resin.

2. Ion Exchange Reaction

- Water flows through a bed of ion exchange resin, where undesirable ions are captured by the resin.
- The resin releases its ions, effectively swapping places with the unwanted ions in the water.

3. Contaminant Removal

- As water passes through the resin bed, contaminants are removed, and treated water exits the system.

4. Resin Regeneration

- Over time, the resin becomes saturated with the exchanged ions and loses its effectiveness.
- Regeneration is performed by flushing the resin with a solution (e.g., brine for cation exchange resins) to restore its ion exchange capacity.

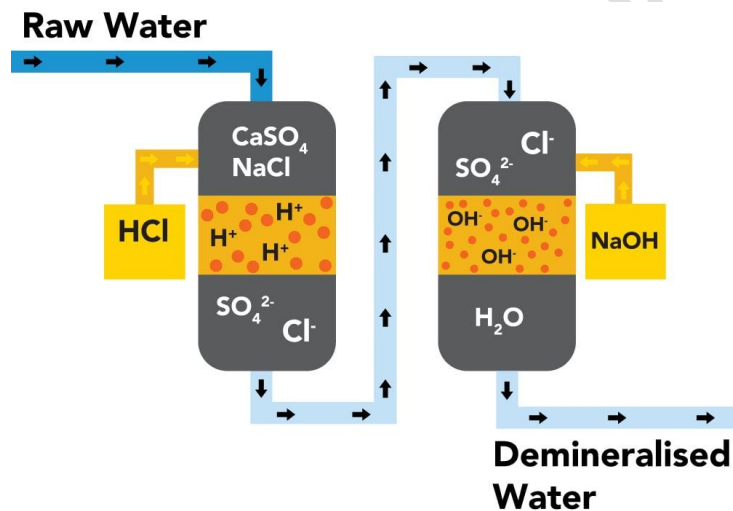
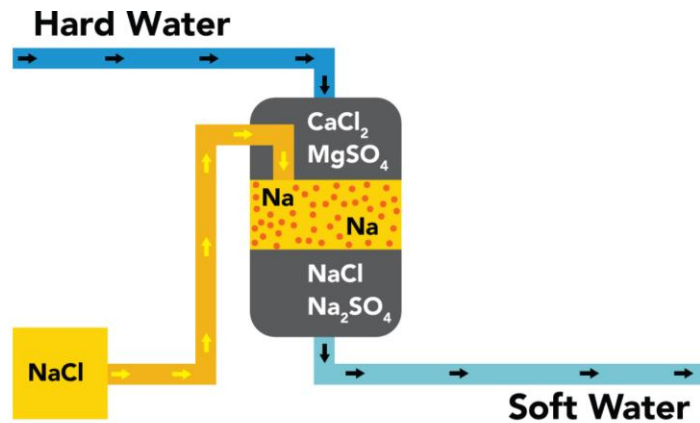
Applications of the Ion Exchange Process in Water Treatment

1. Water Softening

- Hard water contains high levels of calcium and magnesium, which can cause scaling in pipes and appliances.
- The ion exchange process replaces these ions with sodium, preventing scaling and improving water quality.

2. Demineralization

- In industrial processes, the removal of all dissolved salts is often required.
- A combination of cation and anion exchange resins is used to achieve ultrapure water.



3. Nitrate Removal

- The ion exchange process is used to remove nitrates, which can pose health risks, especially to infants.

4. Heavy Metal Removal

- Ion exchange resins can effectively remove heavy metals such as lead, cadmium, and mercury from water, ensuring safety for consumption and environmental compliance.

5. Desalination

- While reverse osmosis is commonly used for seawater desalination, ion exchange is often employed for post-treatment to remove any remaining salts or impurities.

Advantages of the Ion Exchange Process

1. High Efficiency

- The ion exchange process in water treatment achieves precise removal of specific ions, ensuring high-quality water.

2. Versatility

- Applicable for a wide range of water treatment needs, from residential water softening to industrial demineralization.

3. Cost-Effective

- Once installed, ion exchange systems are cost-effective to operate, with minimal energy requirements.

4. Regenerability

- Resins can be regenerated and reused, making the process sustainable over the long term.

5. Compact Design

- Ion exchange systems require relatively little space, making them suitable for installations with limited room.

Disadvantages of ion exchange in water treatment

There are also a few disadvantages to take into account. These include the following:

- They cannot effectively remove bacteria from water
- Long-term costs are expensive
- When ion exchange beds are regenerated, salt water is dumped into the environment

Neutralization

Neutralization is a common practice in wastewater treatment and waste stabilization.

If a waste stream is found to be hazardous because of corrosively, neutralization is the primary treatment used. Moreover, neutralization is used as a pretreatment system before a variety of biological, chemical, and physical treatment processes. Since many chemical treatment processes, such as metal precipitation, coagulation, phosphorus precipitation, and water softening are pH dependent, the pH of these processes is adjusted to achieve maximum process efficiency. Furthermore, the pH of the effluent wastewater from different industrial activities also requires adjustment prior to its discharge into receiving water bodies. The US EPA has set pH standards

for different types of water; for example, the pH range required to protect marine aquatic life is 5–9.

Neutralization is the process of adjusting the pH of water through the addition of an acid or a base, depending on the target pH and process requirements. Some processes such as boiler operations and drinking water standards need neutral water at a pH of 7.

Water or wastewater is generally considered adequately neutralized if (1) its damage to metals, concrete, or other materials is minimal; (2) it has little effect on fish and aquatic life; (3) it has no effect on biological matter (i.e., biological treatment systems).

Parameters

In neutralization, several parameters need to be assessed and evaluated before the actual pH adjustment is carried out. These parameters are discussed in the following sections.

1. pH

pH is the reference indicator for neutralization. Many chemical processes, such as metal precipitation and water softening, which are involved in neutralization, are pH dependent. pH is the negative logarithm of the H^+ ion activity in solution.

$$pH = -\log \{H^+\}$$

In most practical applications, the pH scale ranges from 1 to 14. In pure water and in the absence of materials other than H^+ and OH^- , water behaves ideally and activity equals molar concentration. Under these conditions, $[H^+]$ equals $[OH^-]$ as required by electroneutrality. At 25°C, the ion product of water ($K_w = [H^+][OH^-]$) is 10^{-14} .

The process of neutralization is not only limited to bringing the pH to 7; it is invariably used in the processes, where pH adjustment to other than 7 is required depending on the chemical process in question. For example, some processes like biological wastewater treatment require pH to be near neutral, whereas other processes like metal precipitation require pH to be in the alkaline range. Some of the important chemical processes, where pH plays a significant role and where pH adjustment through neutralization is often required, are metal adsorption and biosorption, chemical precipitation, water softening, coagulation, water fluoridation, and water oxidation.

2. Acidity and Alkalinity

Alkalinity is the capacity of water to neutralize acids, whereas acidity is the capacity of water to neutralize bases. The amount of acid or base to be used in the neutralization process depends upon the respective amount of acidity and alkalinity.

The most important source of both alkalinity and acidity in natural waters is from the carbonate system. However, if the wastewater comes from industrial sources, OH^- or H^+ is also a major contributory factor to alkalinity or acidity, respectively. For example, water from acid mine

drainage contains a large amount of acidity because of the presence of sulfuric acid produced from the oxidation of pyrite. Both acidity and alkalinity are expressed in terms of acid/base equivalents. In water and wastewaters where the predominant ions controlling pH are $[H^+]$, $[OH^-]$, $[HCO_3^-]$, and $[CO_3^{2-}]$, the forms of alkalinity encountered are hydroxide, carbonate, and bicarbonate. These three forms of alkalinity altogether constitute total alkalinity.

3. Buffer Capacity

The word “buffer” stands for the stubbornness against any change. In environmental chemistry, buffers are always defined in the context of pH.

pH buffers are those that resist any changes in solution pH when an acid or a base is added into the solution.

They are very important in chemical neutralization processes. Buffers generally contain a mixture of weak acid and their salts (conjugate base) or weak bases and their conjugate acid. A solution buffered at a particular pH will contain an acid that can react with an externally added base and vice versa. The overall efficiency and chemical cost of the neutralization process depend on the presence of pH buffers in wastewaters.

In natural waters and wastewaters, the buffering capacity arises due to the presence of phosphates, carbonates, and other weak organic acids. The mineral composition of natural waters is regulated by a buffer system involving natural clay minerals such as illite and kaolinite. Careful consideration should be given while neutralizing such waters. If the buffering capacity of the water or wastewater to be neutralized is not taken into account, the actual amount of neutralizing chemical required may vary widely and causes operational problems.

4. Hardness

Hardness in waters arises from the presence of multivalent metallic cations. The principal hardness-causing cations are calcium, magnesium, ferrous iron, and manganous ions. This parameter is important in water-softening processes. The part of the total hardness that is chemically equivalent to the bicarbonate plus carbonate alkalinities is called carbonate hardness.

Process

1. Neutralization of Acidity

The most widely used methods to balance acidity by adding a proper alkaline solution are outlined below:

- Mixing alkaline and acidic wastes such that the net effect is nearly neutral pH.
- Passing the acidic water through a limestone bed. This water should not contain limestone-coating substances such as metal salts or sulfuric or hydrofluoric acids.
- Mixing acid waste with lime slurries or dolomitic slurries.

- Supplementing acidic wastewater with proper amounts of caustic soda or soda ash (Na_2CO_3).

Acidic wastes are neutralized either by adding lime alkalis or by adding sodium alkalis.

The most commonly used lime alkalis are quicklime (CaO) and hydrated or slaked lime ($\text{Ca}(\text{OH})_2$). Sodium alkalis involve the use of caustic soda (NaOH) or soda ash (Na_2CO_3). Calcium and magnesium oxides are considerably less expensive than sodium alkalis and are used more widely. Because these oxides are moderately soluble in water, they are typically slurried. Calcium or magnesium alkalis produce more sludge than do sodium alkalis.

Sodium alkali rapidly reacts with acidic wastes and produces soluble neutral salts when combined with most acidic wastewaters. Between the two types of sodium alkalis, caustic soda is a stronger alkali than soda ash. Caustic soda is available in anhydrous form at various concentrations. Soda ash can be purchased as dry granular material.

Liquid caustic soda is produced and supplied in a concentration range of 50–73%. Most industries use a 50% caustic soda solution. The specific gravity ranges from 1.47 to 1.53 depending on the temperature. Caustic soda is very corrosive in nature. Hence all containers and lines that come in to contact with caustic soda during use or shipment should be carefully selected.

2.Neutralization of Alkalinity

Lowering the pH of a solution is sometimes necessary in some treatment processes or when wastewater is to be discharged in open streams. Discharge of effluent with a pH greater than 8.5 is undesirable and lowering the pH is generally achieved either by adding an acid or by adding carbon dioxide. The process of adding carbon dioxide is called recarbonation and is often practiced in industrial wastewater neutralization. The commonly used acids for pH adjustment of alkaline wastewaters are sulfuric acid (H_2SO_4), hydrochloric acid (HCl), and nitric acid (HNO_3). Among them, sulfuric acid is the most widely used neutralizing agent. Use of nitric acid is restricted because of more stringent nutrient effluent limitations. There is no direct relationship between pH and alkalinity.

Hence, titration curves should be established in laboratories before the design of an alkaline wastewater neutralization system. Sulfuric acid used in wastewater treatment could be 77.7% concentration or 97% concentration with an approximate specific gravity of 1.83. Sulfuric acid releases a significant amount of heat when added to water. Precautionary measures must be taken to avoid any chemical accident due to the heat generated when practicing neutralization with sulfuric acid. Hydrochloric acid has an average specific gravity of 1.17 and an acid content of 33% by weight. Properly lined tanks should be used to store this classification of hydrochloric acid. Generally polyvinyl chloride tanks or lined steel tanks are used.

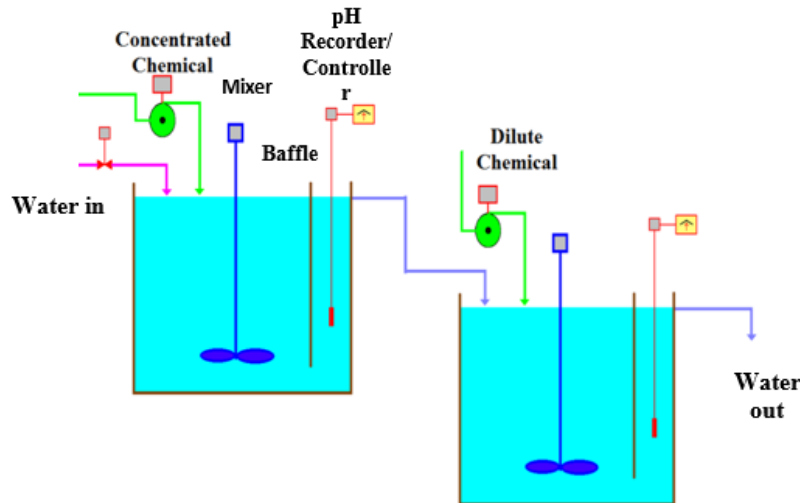


Fig 2 Continuous Neutralization System in two stages

Oxidation

Oxidation in water treatment refers to a process in which oxidizing agents interact with contaminants to break them down into harmless compounds. This method is particularly effective for treating industrial effluents, hazardous organic compounds, and even certain metals. The oxidation process can be categorized into two main types:

1. Biological Oxidation: Utilizes microorganisms to break down organic pollutants in wastewater.
2. Chemical Oxidation: Uses strong oxidizing chemicals to degrade pollutants and disinfect water.

Among these, chemical oxidation in wastewater treatment is the most effective for handling complex pollutants that biological methods may not fully degrade.

Chemical Oxidation in Wastewater Treatment: How It Works

Chemical oxidation in wastewater treatment involves the addition of strong oxidants to wastewater, which react with contaminants to form less harmful substances. The most commonly used oxidants include:

- Chlorine and Hypochlorite: Effective in disinfection and oxidation of organic matter.
- Ozone (O₃): A powerful oxidant that rapidly breaks down pollutants without leaving harmful residues.
- Hydrogen Peroxide (H₂O₂): Often used in combination with other oxidants for enhanced treatment.

- Potassium Permanganate (KMnO_4): Applied in water treatment for removing iron, manganese, and organic contaminants.
- Fenton's Reagent ($\text{H}_2\text{O}_2 + \text{Fe}^{2+}$): A strong oxidation process that generates hydroxyl radicals for the breakdown of persistent pollutants.

Benefits of Oxidation Water Treatment

1. Efficient Removal of Contaminants

- Eliminates organic pollutants, heavy metals, and pathogens.
- Reduces chemical oxygen demand (COD) and biological oxygen demand (BOD), improving water quality.

2. Disinfection and Pathogen Control

- Kills bacteria, viruses, and other microorganisms without harmful residuals.
- Prevents the spread of waterborne diseases.

3. Enhanced Water Clarity and Aesthetics

- Removes color, odor, and turbidity from wastewater.
- Improves the overall aesthetic appeal of treated water.

4. Compatibility with Other Treatment Methods

- Can be integrated with biological treatment, filtration, and advanced oxidation processes (AOPs).
- Works efficiently in combination with coagulants and flocculants.

Challenges and Future Developments

While oxidation water treatment is highly effective, certain challenges remain:

- High Operational Costs: Some oxidation methods require significant energy input, increasing treatment costs.
- Formation of Byproducts: Some oxidants produce undesirable byproducts that must be managed carefully.
- Handling of Strong Oxidants: Certain chemicals used in oxidation require special handling and storage precautions.

Reduction

In water treatment, chemical reduction is used for highly specific tasks: eliminating residual oxidants (like chlorine), removing dissolved oxygen to prevent system corrosion, and converting toxic hexavalent chromium (Cr^{VI}) into less toxic trivalent chromium (Cr^{III}).

Key Reduction Processes

1. Dechlorination (Elimination of Residual Oxidants)

- **Purpose:** Removes excess chlorine or chloramines from drinking water or reverse osmosis feed streams to protect membranes from degradation and eliminate taste and odor issues.
- **Chemical Agents:** Sulfur dioxide (SO_2) gas, sodium bisulfite (NaHSO_3), or sodium metabisulfite ($\text{Na}_2\text{S}_2\text{O}_5$).
- **Reaction Example:** When sodium bisulfite reacts with free chlorine (HOCl), it reduces the chlorine to chloride while neutralizing the oxidants:
$$\text{HOCl} + \text{NaHSO}_3 \rightarrow \text{HCl} + \text{NaHSO}_4$$

2. Chromium Reduction (Heavy Metal Detoxification)

- **Purpose:** Converts highly toxic, soluble hexavalent chromium (Cr^{VI}) into insoluble, less toxic trivalent chromium (Cr^{III}), which can then be precipitated out of industrial wastewater.
- **Chemical Agents:** Sulfur dioxide, sodium metabisulfite, or ferrous sulfate (FeSO_4) under acidic conditions (typically $\text{pH} < 3.0$).
- **Reaction Example:** Using sulfur dioxide in an acidic medium:
$$2\text{H}_2\text{CrO}_4 + 3\text{SO}_2 \rightarrow \text{Cr}_2(\text{SO}_4)_3 + 2\text{H}_2\text{O}$$

3. Deoxygenation (Corrosion Control)

- **Purpose:** Removes dissolved oxygen (O_2) from industrial process water and boiler feed water to prevent oxidative pipe corrosion.
- **Chemical Agents:** Hydrazine (N_2H_4), sodium sulfite (Na_2SO_3), or catalyzed reducing agents.
- **Reaction Example:** Sodium sulfite reacting with dissolved oxygen:
$$2\text{Na}_2\text{SO}_3 + \text{O}_2 \rightarrow 2\text{Na}_2\text{SO}_4$$